



basic education

Department:
Basic Education
REPUBLIC OF SOUTH AFRICA

SENIOR CERTIFICATE EXAMINATIONS/ NATIONAL SENIOR CERTIFICATE EXAMINATIONS

PHYSICAL SCIENCES: PHYSICS (P1)

MAY/JUNE 2026

MARKS: 150

TIME: 3 hours

This question paper consists of 17 pages and 3 data sheets.

INSTRUCTIONS AND INFORMATION

1. Write your centre number and examination number in the appropriate spaces on the ANSWER BOOK.
2. This question paper consists of TEN questions. Answer ALL the questions in the ANSWER BOOK.
3. Start EACH question on a NEW page in the ANSWER BOOK.
4. Number the answers correctly according to the numbering system used in this question paper.
5. Leave ONE line between two subquestions, e.g. between QUESTION 2.1 and QUESTION 2.2.
6. You may use a non-programmable calculator.
7. Show ALL formulae and substitutions in ALL calculations.
8. Round off your final numerical answers to a minimum of TWO decimal places.
9. Give brief motivations, discussions etc. where required.
10. You are advised to use the attached DATA SHEETS.
11. Write neatly and legibly.

QUESTION 1: MULTIPLE-CHOICE QUESTIONS

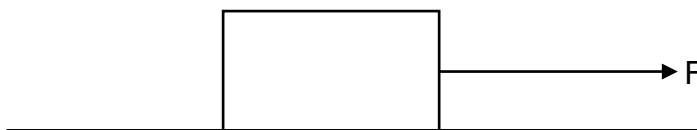
Various options are provided as possible answers to the following questions. Each question has only ONE correct answer. Choose the answer and write only the letter (A–D) next to the question numbers (1.1 to 1.10) in the ANSWER BOOK, e.g. 1.11 E.

- 1.1 The gravitational force experienced by an object on the surface of a planet is inversely proportional to the ...

- A mass of the planet.
- B mass of the object.
- C square of the radius of the planet.
- D radius of the planet.

(2)

- 1.2 A horizontal force F acts on a block placed on a rough horizontal surface. The block, starting from rest, moves to the right at constant acceleration.



Which ONE of the following statements is CORRECT?

- A F is equal to the kinetic frictional force.
- B The momentum of the block increases.
- C The net work done on the block is zero.
- D The normal force is less than the weight of the block.

(2)

- 1.3 Objects A and B are thrown vertically upwards with velocities v and $2v$ respectively. Object A reaches a maximum height h .

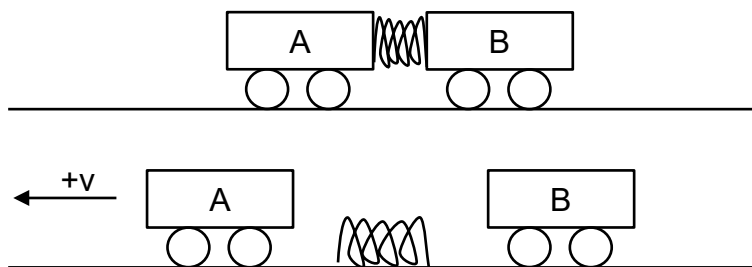
What is the maximum height reached by object B?

Ignore the effects of friction.

- A h
- B $2h$
- C $\frac{1}{2}h$
- D $4h$

(2)

- 1.4 Two trolleys A and B, of masses m_1 and m_2 respectively, are held at rest on a frictionless surface with a compressed spring between them.



When the trolleys are released, the spring expands and drops. If the velocity of trolley A is now $+v$, what is the velocity of trolley B?

A $+\frac{m_1}{m_2}v$

B $-\frac{m_1}{m_2}v$

C $+\frac{m_2}{m_1}v$

D $-\frac{m_2}{m_1}v$

(2)

- 1.5 A motor lifts a load vertically upwards at a constant speed.

Which ONE of the following combinations of the work done by the motor and the power dissipated by the motor, is CORRECT?

	WORK DONE BY THE MOTOR	POWER DISSIPATED BY THE MOTOR
A	Constant	Increases
B	Increases	Increases
C	Increases	Constant
D	Constant	Constant

(2)

- 1.6 The ratio of the wavelength of sound produced by a stationary source (λ_s) to the wavelength of the sound detected by a listener (λ_L) is 1 : 1,09. The listener is moving at constant velocity.

The frequency of the source is f_s and the frequency detected by the listener is f_L .

Which ONE of the following combinations of f_L and the movement of the listener is CORRECT?

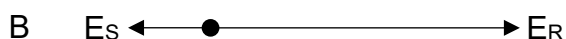
	f_L	MOVEMENT OF THE LISTENER
A	Greater than f_s	Towards the source
B	Greater than f_s	Away from the source
C	Less than f_s	Towards the source
D	Less than f_s	Away from the source

(2)

- 1.7 R and S are two NEGATIVE point charges of equal magnitude. P is a point to the left of R. Point P and charges R and S are on the same horizontal line.

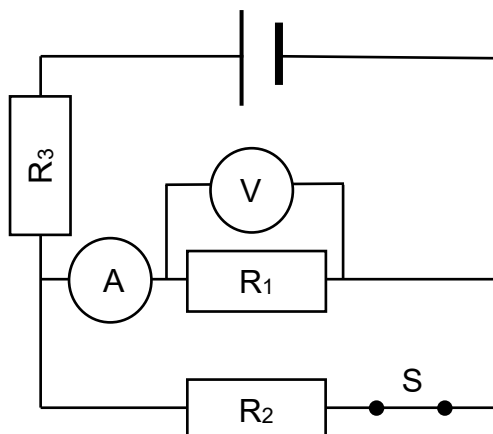


Which ONE of the following vector diagrams, drawn to scale, of the electric fields E_R due to R, and E_S due to S, at point P, is CORRECT?



(2)

- 1.8 The battery in the circuit below has negligible internal resistance. Switch S is closed.



Switch S is now opened. How will the readings on the ammeter and voltmeter be affected?

	AMMETER READING	VOLTMETER READING
A	Decreases	Increases
B	Increases	Decreases
C	Decreases	Decreases
D	Increases	Increases

(2)

- 1.9 The magnitude of the induced emf in an AC generator can be increased by:

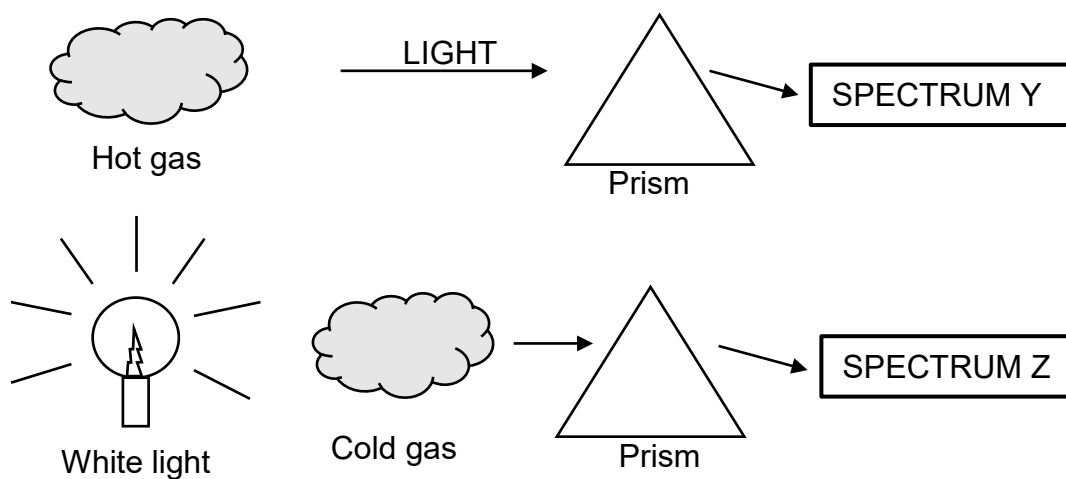
- (i) Increasing the strength of the magnetic field
- (ii) Increasing the speed at which the coil is rotated
- (iii) Increasing the number of turns in the coil

Which of the statements above are CORRECT?

- A (i) and (ii) only
- B (i) and (iii) only
- C (ii) and (iii) only
- D (i), (ii) and (iii)

(2)

- 1.10 The simplified diagrams below show how different types of spectra are obtained.



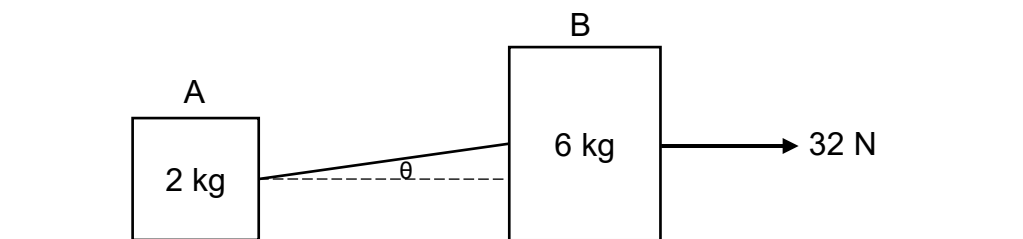
Which ONE of the following statements is CORRECT?

- A Y is an emission spectrum and Z is an absorption spectrum.
- B Y is an emission spectrum and Z is a continuous spectrum.
- C Y is a continuous spectrum and Z is an absorption spectrum.
- D Y is an absorption spectrum and Z is an emission spectrum.

(2)
[20]

QUESTION 2 (Start on a new page.)

Two blocks A and B, of masses 2 kg and 6 kg respectively, are placed on a rough horizontal surface and are connected by a light, inextensible string. The string makes an angle θ with the horizontal.



When a horizontal force of 32 N is applied to block B, the blocks accelerate to the right at $0,47 \text{ m}\cdot\text{s}^{-2}$. Block A now experiences a constant frictional force of 3,74 N.

2.1 State *Newton's Second Law of Motion* in words. (2)

2.2 Draw a labelled free-body diagram showing ALL the forces acting on block B. (5)

2.3 Calculate:

2.3.1 The magnitude of the frictional force acting on block B by applying Newton's Second Law to each block separately (5)

2.3.2 θ , if the coefficient of kinetic friction for block A and the surface is 0,2 (5)

2.4 As the blocks move, the string breaks. Will the kinetic frictional force experienced by block A now be GREATER THAN, LESS THAN or EQUAL TO 3,74 N?

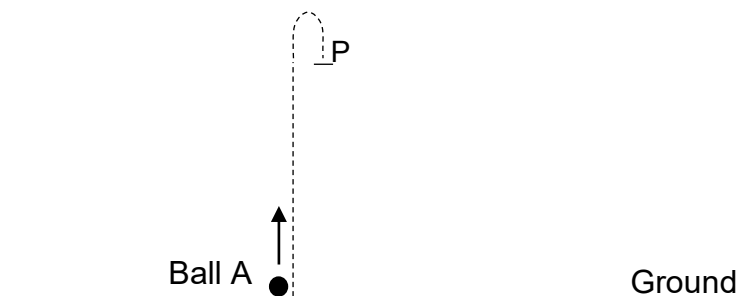
Explain the answer without the use of a calculation.

(3)
[20]

QUESTION 3 (Start on a new page.)

Ball A is projected vertically upwards at a velocity of $20 \text{ m}\cdot\text{s}^{-1}$ from the ground. On its way down, it passes point P with a speed of $6,87 \text{ m}\cdot\text{s}^{-1}$.

Ignore the effects of friction.



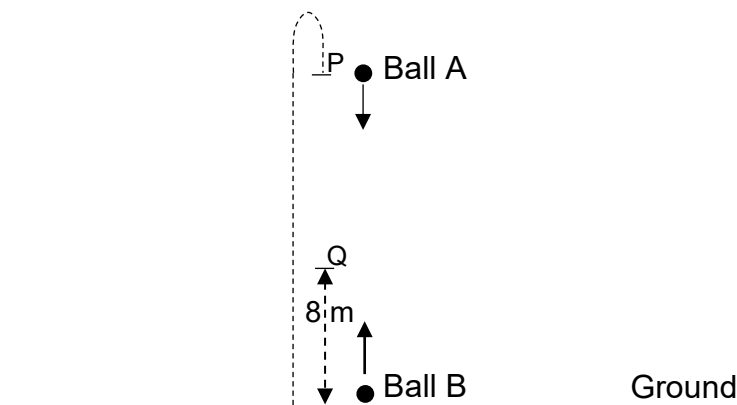
3.1 Define the term *projectile*. (2)

3.2 Using EQUATIONS OF MOTIONS ONLY, calculate the:

3.2.1 Time taken for ball A to reach point P on its way down (3)

3.2.2 The height of point P above the ground (3)

When ball A reaches point P, ball B is projected vertically upwards from the ground. The balls pass each other at point Q which is 8 m above the ground.



3.3 Calculate the speed at which ball B was projected. (5)

3.4 On the same set of axes, sketch the position versus time graph for ball A and ball B from the time ball A was projected until both reach point Q. Take the ground as the zero position. Label the graph for ball A and ball B as A and B respectively.

Indicate the following numerical values on the graph:

- Time taken for ball A to reach point P on its way down
- Height of point P
- Height of point Q

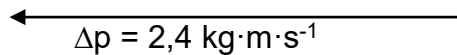
(5)
[18]

QUESTION 4 (Start on a new page.)

Ball A and ball B move towards each other along a horizontal surface. Ball B has a mass of 0,3 kg and an initial speed of $15 \text{ m}\cdot\text{s}^{-1}$. The balls collide.

The vector shown below represents the change in momentum of ball A.

Ignore the effects of friction.

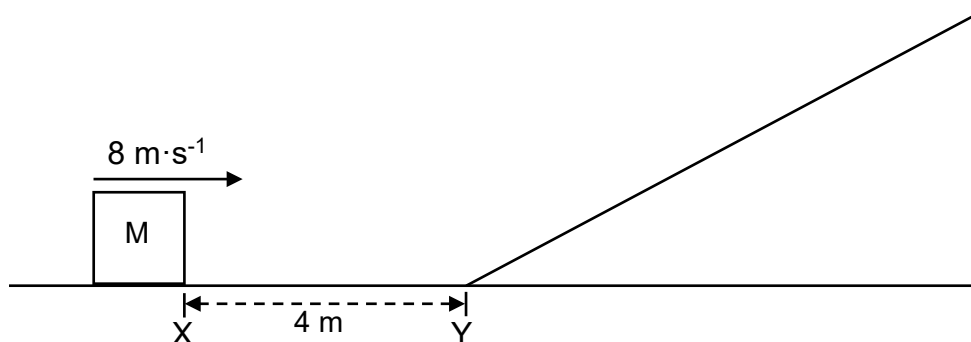

$$\Delta p = 2,4 \text{ kg}\cdot\text{m}\cdot\text{s}^{-1}$$

- 4.1 State the *principle of conservation of linear momentum* in words. (2)
- 4.2 What was the initial direction of ball A? Choose from LEFT or RIGHT. (2)
- 4.3 Calculate the velocity of ball B after the collision. (5)

[9]

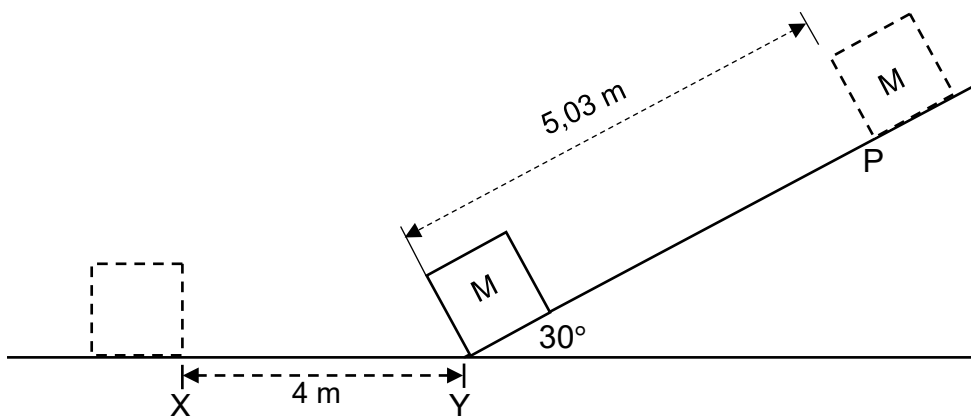
QUESTION 5 (Start on a new page.)

A block of mass M kg moves along a rough horizontal surface while experiencing a constant frictional force of 3 N . It passes point X with a speed of $8\text{ m}\cdot\text{s}^{-1}$ and reaches point Y with a speed v_Y . Point Y is 4 m to the right of point X.



- 5.1 Using ENERGY PRINCIPLES ONLY, show that v_Y^2 is given by $v_Y^2 = 64 - \frac{24}{M}$. (3)

The block then moves up an incline that is at an angle of 30° with the horizontal. The block experiences a constant frictional force of 2 N as it moves a distance of $5,03\text{ m}$ up the incline and comes to rest at point P.



- 5.2 USING ENERGY PRINCIPLES ONLY, calculate the mass of the block. (5)
[8]

QUESTION 6 (Start on a new page.)

- 6.1 A stationary sound source (S) produces sound with a frequency of 170 Hz. A listener (L) in a car travelling at a constant speed of $12 \text{ m}\cdot\text{s}^{-1}$ towards this source, observes the frequency of the sound as x Hz.

Take the speed of sound in air to be $340 \text{ m}\cdot\text{s}^{-1}$.



- 6.1.1 State the *Doppler effect* in words. (2)

- 6.1.2 Will the observed frequency INCREASE, DECREASE or REMAIN THE SAME if:

- (a) The constant speed of the car moving towards the stationary sound source is greater than $12 \text{ m}\cdot\text{s}^{-1}$ (1)
- (b) The sound produced by the source is louder (1)

The frequency of the sound source is changed. The listener now travels towards the stationary sound source at a constant speed of $24 \text{ m}\cdot\text{s}^{-1}$.

The observed frequency is still x Hz.



- 6.1.3 Calculate the new frequency of the sound source. (6)

- 6.2 **X**, **Y** and **Z** are absorption spectra. B and R represent the blue and red ends of the spectra. Spectrum **X** shows the absorption lines of light from the sun.

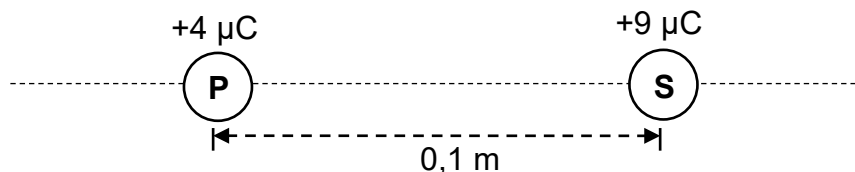


Which spectrum shows the absorption lines of light from another star that is moving AWAY from Earth? Choose from **X**, **Y** or **Z**. Explain the answer.

(3)
[13]

QUESTION 7 (Start on a new page.)

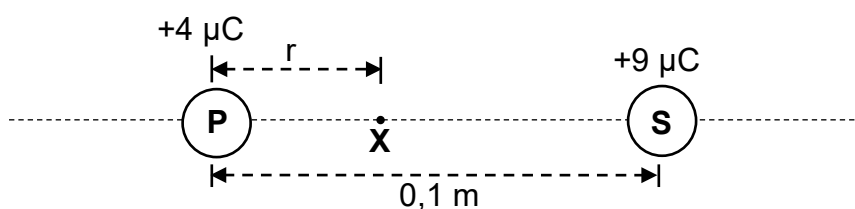
Two small spheres **P** and **S**, carrying charges of $+4\ \mu\text{C}$ and $+9\ \mu\text{C}$ respectively, are placed $0,1\ \text{m}$ apart.



7.1 State *Coulomb's law* in words. (2)

7.2 Calculate the magnitude of the electrostatic force experienced by sphere **S**. (3)

X is a point r metres to the right of sphere **P**, as shown in the diagram below.



7.3 Define the term *electric field at a point*. (2)

7.4 Calculate r if the net electric field at point **X** is zero. (5)

7.5 The two spheres are brought into contact and returned to their original positions.

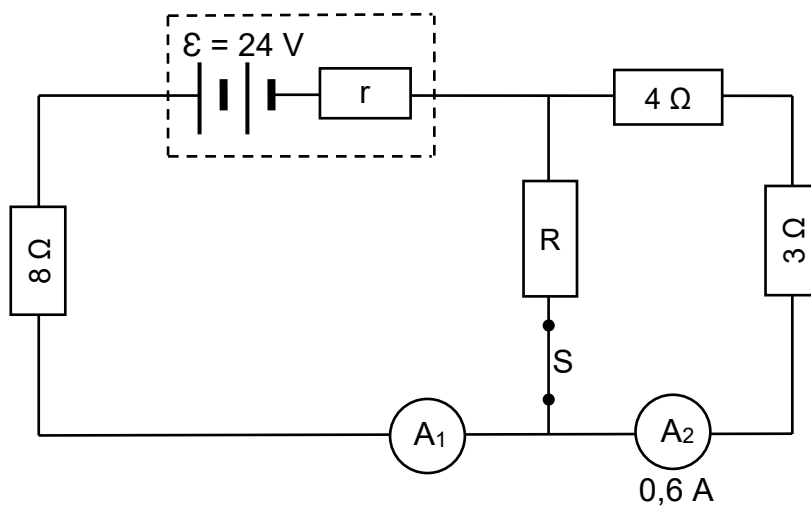
7.5.1 Were electrons transferred from **sphere P to S** or from **sphere S to P** when they were in contact? (1)

7.5.2 At what distance from sphere **P** will the net electric field now still be zero? (1)

[14]

QUESTION 8 (Start on a new page.)

- 8.1 A battery with an emf of 24 V and an unknown internal resistance (r) is connected to four resistors. When switch S is closed, the reading on ammeter A_2 is 0,6 A.



- 8.1.1 Explain what is meant by an *emf* of 24 V. (2)

The power dissipated by the 8 Ω resistor is 46,08 W.

- 8.1.2 Calculate the resistance of resistor R. (6)

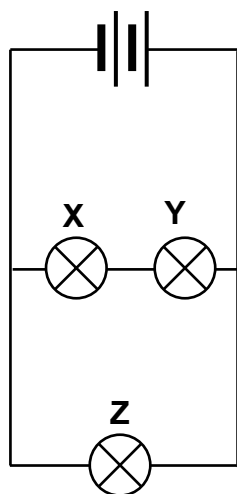
- 8.1.3 Calculate the internal resistance (r) of the battery. (5)

- 8.1.4 How does the potential difference across the 4 Ω resistor compare to that across resistor R? Choose from GREATER THAN, LESS THAN or EQUAL TO. Give a reason for the answer. (2)

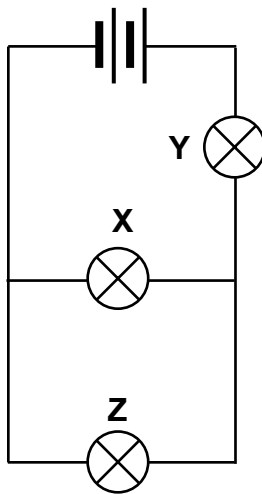
Switch S is now opened.

- 8.1.5 How will the power dissipated by the 8 Ω resistor now be affected? Choose from INCREASES, DECREASES or REMAINS THE SAME. (1)

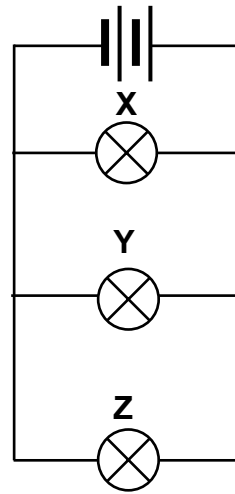
- 8.2 Different arrangements of three identical light bulbs **X**, **Y** and **Z** are shown in circuits A, B and C below. The batteries are identical with negligible internal resistance.



CIRCUIT A



CIRCUIT B

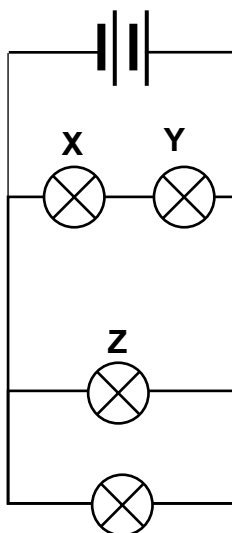


CIRCUIT C

- 8.2.1 In which circuit will bulb **X** burn the brightest? Choose from A, B or C. Explain the answer.

(3)

In circuit A, an additional identical light bulb is now connected in parallel to bulb **Z**. This is shown below.

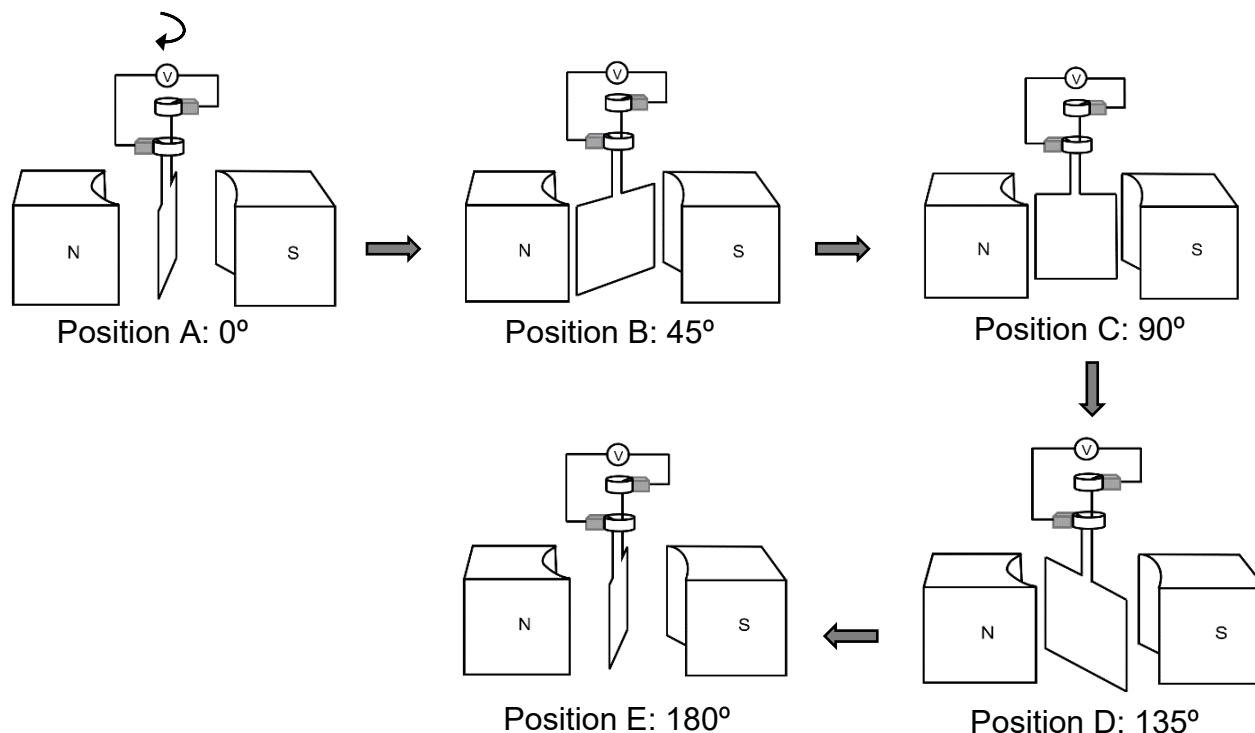


- 8.2.2 How will the brightness of bulb **X** now be affected? Choose from INCREASES, DECREASES or REMAINS THE SAME. Give a reason for the answer.

(2)
[21]

QUESTION 9 (Start on a new page.)

The diagrams below show different positions of the coil, rotating in a clockwise direction, in an AC generator.



- 9.1 Use the principle of electromagnetic induction to explain how an AC generator works. (2)
- 9.2 At which position(s) during the rotation of the coil will a maximum emf be induced? (2)
- 9.3 Draw a graph of the induced emf versus time. Indicate the positions A, B, C, D and E on the graph. (3)
- 9.4 A $100\ \Omega$ resistor is connected to this generator. If the peak voltage produced by the generator is 170 V, calculate the rms current passing through the resistor. (4)
- 9.5 This generator is now changed to a DC generator. The coil is rotated at the same speed as before.
- 9.5.1 What change was made to the AC generator to convert it to a DC generator? (1)
- 9.5.2 State the energy conversion in a DC generator. (1)
- 9.5.3 Will the peak voltage of this generator still be 170 V? Choose from YES or NO. (1)

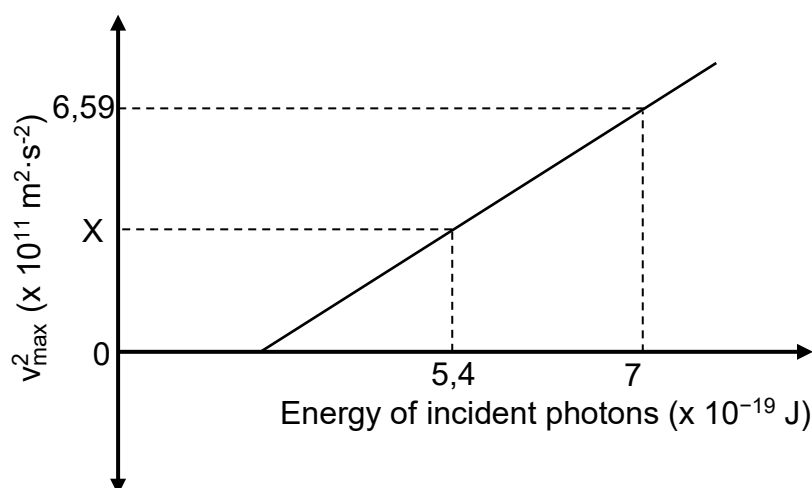
[14]

QUESTION 10 (Start on a new page.)

In an experiment, a metal is identified using the photoelectric effect. Photons of light with different energies are shone onto the metal surface and the corresponding maximum velocities of the ejected electrons are recorded.

The graph below shows the relationship between the square of the maximum velocities of the ejected electrons, $v_{\max}^2$, and the energy of the incident photons.

The value of X is unknown.



10.1 Define the term *work function* of a metal. (2)

One of the metals in the table below was used in this experiment.

METAL	WORK FUNCTION (x 10^{-19} J)
Barium	4
Chromium	5
Aluminium	7

10.2 Use a suitable calculation to identify the metal used. (4)

10.3 Calculate X. (5)

10.4 How will the maximum velocities of the ejected electrons change when the intensity of the incident light is increased? Choose from INCREASES, DECREASES or REMAINS THE SAME. Give a reason for the answer. (2)
[13]

TOTAL: 150

**DATA FOR PHYSICAL SCIENCES GRADE 12
PAPER 1 (PHYSICS)**

**GEGEWENS VIR FISIIESE WETENSKAPPE GRAAD 12
VRAESTEL 1 (FISIKA)**

TABLE 1: PHYSICAL CONSTANTS/TABEL 1: FISIIESE KONSTANTES

NAME/NAAM	SYMBOL/SIMBOOL	VALUE/WAARDE
Acceleration due to gravity <i>Swaartekragversnelling</i>	g	$9,8 \text{ m} \cdot \text{s}^{-2}$
Universal gravitational constant <i>Universele gravitasiekonstante</i>	G	$6,67 \times 10^{-11} \text{ N} \cdot \text{m}^2 \cdot \text{kg}^{-2}$
Radius of the Earth <i>Radius van die Aarde</i>	R_E	$6,38 \times 10^6 \text{ m}$
Mass of the Earth <i>Massa van die Aarde</i>	M_E	$5,98 \times 10^{24} \text{ kg}$
Speed of light in a vacuum <i>Spoed van lig in 'n vakuum</i>	c	$3,0 \times 10^8 \text{ m} \cdot \text{s}^{-1}$
Planck's constant <i>Planck se konstante</i>	h	$6,63 \times 10^{-34} \text{ J} \cdot \text{s}$
Coulomb's constant <i>Coulomb se konstante</i>	k	$9,0 \times 10^9 \text{ N} \cdot \text{m}^2 \cdot \text{C}^{-2}$
Charge on electron <i>Lading op elektron</i>	e	$1,6 \times 10^{-19} \text{ C}$
Electron mass <i>Elektronmassa</i>	m_e	$9,11 \times 10^{-31} \text{ kg}$

TABLE 2: FORMULAE/TABEL 2: FORMULES**MOTION/BEWEGING**

$v_f = v_i + a\Delta t$	$\Delta x = v_i\Delta t + \frac{1}{2}a\Delta t^2$ or/of $\Delta y = v_i\Delta t + \frac{1}{2}a\Delta t^2$
$v_f^2 = v_i^2 + 2a\Delta x$ or/of $v_f^2 = v_i^2 + 2a\Delta y$	$\Delta x = \left(\frac{v_i + v_f}{2}\right)\Delta t$ or/of $\Delta y = \left(\frac{v_i + v_f}{2}\right)\Delta t$

FORCE/KRAG

$F_{\text{net}} = ma$	$p = mv$
$f_s^{\text{max}} = \mu_s N$	$f_k = \mu_k N$
$F_{\text{net}}\Delta t = \Delta p$ $\Delta p = mv_f - mv_i$	$w = mg$
$F = G \frac{m_1 m_2}{d^2}$ or/of $F = G \frac{m_1 m_2}{r^2}$	$g = G \frac{M}{d^2}$ or/of $g = G \frac{M}{r^2}$

WORK, ENERGY AND POWER/ARBEID, ENERGIE EN DRYWING

$W = F\Delta x \cos \theta$	$U = mgh$ or/of $E_p = mgh$
$K = \frac{1}{2}mv^2$ or/of $E_k = \frac{1}{2}mv^2$	$W_{\text{net}} = \Delta K$ or/of $W_{\text{net}} = \Delta E_k$ $\Delta K = K_f - K_i$ or/of $\Delta K = E_{kf} - E_{ki}$
$W_{\text{nc}} = \Delta K + \Delta U$ or/of $W_{\text{nc}} = \Delta E_k + \Delta E_p$	$P = \frac{W}{\Delta t}$
$P_{\text{ave}} = Fv_{\text{ave}}$ / $P_{\text{gemid}} = Fv_{\text{gemid}}$	

WAVES, SOUND AND LIGHT/GOLWE, KLANK EN LIG

$v = f\lambda$	$T = \frac{1}{f}$
$f_L = \frac{v \pm v_L}{v \pm v_s} f_s$ / $f_L = \frac{v \pm v_L}{v \pm v_b} f_b$	$E = hf$ or/of $E = \frac{hc}{\lambda}$
$E = W_0 + E_{k(\text{max})}$ or/of $E = W_0 + K_{\text{max}}$ where/waar	
$E = hf$ and/en $W_0 = hf_0$ and/en $E_{k(\text{max})} = \frac{1}{2}mv_{\text{max}}^2$ or/of $K_{\text{max}} = \frac{1}{2}mv_{\text{max}}^2$	

ELECTROSTATICS/ELEKTROSTATIKA

$F = \frac{kQ_1Q_2}{r^2}$	$E = \frac{kQ}{r^2}$
$V = \frac{W}{q}$	$E = \frac{F}{q}$
$n = \frac{Q}{e} \quad \text{or/of} \quad n = \frac{Q}{q_e}$	

ELECTRIC CIRCUITS/ELEKTRIESE STROOMBANE

$R = \frac{V}{I}$	emf ($\mathcal{E}$) = I(R + r) emk ($\mathcal{E}$) = I(R + r)
$R_s = R_1 + R_2 + \dots$ $\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$	$q = I\Delta t$
$W = Vq$ $W = VI\Delta t$ $W = I^2R\Delta t$ $W = \frac{V^2\Delta t}{R}$	$P = \frac{W}{\Delta t}$ $P = VI$ $P = I^2R$ $P = \frac{V^2}{R}$

ALTERNATING CURRENT/WISSELSTROOM

$I_{\text{rms}} = \frac{I_{\text{max}}}{\sqrt{2}} \quad / \quad I_{\text{wgk}} = \frac{I_{\text{maks}}}{\sqrt{2}}$ $V_{\text{rms}} = \frac{V_{\text{max}}}{\sqrt{2}} \quad / \quad V_{\text{wgk}} = \frac{V_{\text{maks}}}{\sqrt{2}}$	$P_{\text{ave}} = V_{\text{rms}} I_{\text{rms}} \quad / \quad P_{\text{gemid}} = V_{\text{wgk}} I_{\text{wgk}}$ $P_{\text{ave}} = I_{\text{rms}}^2 R \quad / \quad P_{\text{gemid}} = I_{\text{wgk}}^2 R$ $P_{\text{ave}} = \frac{V_{\text{rms}}^2}{R} \quad / \quad P_{\text{gemid}} = \frac{V_{\text{wgk}}^2}{R}$
--	---